

PSS Week 12 P12

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1 Original Statement

Denote by W_1, W_2, W_3, W_4 the four witnesses. Let $S_1 = S_2 = \{1, 2\}$. If W_1 and W_2 have the same reply, then the liar lies among W_3 and W_4 . First assume that W_1 and W_2 both said yes, so ask the naive $S_3 = \{1, 2\}$. If W_3 says no, they are the liar. Otherwise, W_4 is the liar. Either way, we can take $S_4 = \{1\}$ and determine the murderer. The case where W_1 and W_2 said no is the same, but we instead ask W_3 and W_4 questions on $\{3, 4\}$.

Now suppose W_1 and W_2 disagree, so that one of them is the liar. The two remaining W_3 and W_4 must be truth tellers, so we can perform binary search: First ask $S_3 = \{1, 2\}$. If yes, then take $S_4 = \{1\}$ and determine the murderer. Otherwise, take $S_4 = \{3\}$ and finish.